

ARIMA Methodology - Comprehensive Guide

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ARIMA Methodology - Comprehensive Guide

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1. Introduction to Time Series Forecasting

What is Time Series Data?

Time series data consists of observations recorded sequentially at regular intervals over time. Examples include:

- Daily stock prices
- Monthly sales figures
- Hourly temperature readings
- Quarterly GDP data

Why Forecast?

Forecasting time series helps:

- ****Decision Making:**** Plan inventory, staffing, resources

- **Risk Management:** Identify potential threats
- **Strategy:** Develop long-term business plans
- **Investment:** Optimize portfolio allocation

ARIMA Overview

ARIMA (AutoRegressive Integrated Moving Average) is a statistical method for time series forecasting that:

- Captures temporal dependencies
- Handles trending and seasonal patterns
- Provides confidence intervals
- Requires minimal external data

2. Stationary vs Non-Stationary Series

Stationarity Definition

A time series is **stationary** if its statistical properties (mean, variance, autocorrelation) remain constant over time.

Mathematical Definition:

For a stationary process Y_t :

- $E[Y_t] = \mu$ (constant mean)
- $\text{Var}(Y_t) = \sigma^2$ (constant variance)
- $\text{Cov}(Y_t, Y_{t-h})$ depends only on h , not on t

Why Stationarity Matters

Most ARIMA methods assume stationarity. Non-stationary series:

- Have time-dependent mean or variance
- Cannot be reliably forecast
- May produce spurious relationships

Testing for Stationarity

1. Augmented Dickey-Fuller (ADF) Test

Null Hypothesis (H_0): Series has a unit root (non-stationary)

Alternative (H_1): Series is stationary

Interpretation:

- $p\text{-value} < 0.05$: Reject $H_0 \rightarrow$ Series is stationary ✓
- $p\text{-value} \geq 0.05$: Fail to reject $H_0 \rightarrow$ Series is non-stationary ✗

Example:

```

ADF Test Results:

Statistic: -2.45

p-value: 0.127  
Result: Non-stationary ( $p > 0.05$ )  
Recommendation: Apply differencing ( $d=1$ )  
...

## #### 2. KPSS Test

**\*\*Null Hypothesis (H0):\*\*** Series is stationary  
**\*\*Alternative (H1):\*\*** Series has a unit root

**\*\*Interpretation:\*\***

- p-value  $> 0.05$ : Fail to reject  $H_0 \rightarrow$  Series is stationary ✓
- p-value  $\leq 0.05$ : Reject  $H_0 \rightarrow$  Series is non-stationary ✗

## ### Making Series Stationary

### #### Differencing

Transform the series by taking differences:

$$\Delta y_t = y_t - y_{t-1}$$

**\*\*First Differencing ( $d=1$ ):\*\***  
...

Original: [100, 110, 108, 112]

First Diff: [10, -2, 4]  
...

**\*\*Second Differencing ( $d=2$ ):\*\***  
...

First Diff: [10, -2, 4]

Second Diff: [-12, 6]  
...

### #### Seasonal Differencing

For series with seasonal patterns (e.g., 12-month cycle):

$$\Delta_s y_t = y_t - y_{t-s}$$

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## ## 3. ARIMA Components

### ### AutoRegressive (AR) Component - Parameter $p$

AR models express the current value as a linear function of past values:

**\*\*AR(1):\*\***  $y_t = \phi_1 \cdot y_{t-1} + \epsilon_t$

**\*\*AR(p):\*\***  $y_t = \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \dots + \phi_p \cdot y_{t-p} + \epsilon_t$

**\*\*Interpretation:\*\***

- p: Number of past observations to use
- Higher p: Longer memory of past values
- Typical range:  $p \in [0, 5]$

**\*\*When to use:\*\***

- When current value depends on recent history
- When ACF dies down gradually

**### Integrated (I) Component - Parameter d**

Differencing order to achieve stationarity:

- d=0: Series is already stationary, no differencing
- d=1: Apply first differencing once
- d=2: Apply first differencing twice (rare)

**\*\*Guide:\*\***

- Check stationarity tests
- Apply minimum differencing needed
- Typical range:  $d \in [0, 2]$

**### Moving Average (MA) Component - Parameter q**

MA models express the current value as a linear combination of past forecast errors:

**\*\*MA(1):\*\***  $y_t = \varepsilon_t + \theta_1 \cdot \varepsilon_{t-1}$

**\*\*MA(q):\*\***  $y_t = \varepsilon_t + \theta_1 \cdot \varepsilon_{t-1} + \theta_2 \cdot \varepsilon_{t-2} + \dots + \theta_q \cdot \varepsilon_{t-q}$

**\*\*Interpretation:\*\***

- q: Number of lag terms to use
- Higher q: Longer "memory" of errors
- Typical range:  $q \in [0, 5]$

**\*\*When to use:\*\***

- When shocks have temporary effects
- When ACF cuts off after lag q

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**## 4. ARIMA Model Selection**

**### ACF and PACF Analysis**

**#### ACF (AutoCorrelation Function)**

Measures correlation between observations at different lags.

**\*\*Interpretation:\*\***

- AR(p) series: PACF cuts off at lag p, ACF dies gradually
- MA(q) series: ACF cuts off at lag q, PACF dies gradually
- ARMA(p,q) series: Both ACF and PACF die gradually

### #### PACF (Partial AutoCorrelation Function)

Measures direct correlation between observations, removing intermediate effects.

#### \*\*Usage Guide:\*\*

| ACF Pattern    | PACF Pattern   | Suggested Model |
|----------------|----------------|-----------------|
| Dies slowly    | Cuts at lag p  | AR(p)           |
| Cuts at lag q  | Dies slowly    | MA(q)           |
| Dies gradually | Dies gradually | ARMA(p,q)       |
| Cuts at lag 1  | Cuts at lag 1  | ARMA(1,1)       |

### ### Automatic Parameter Selection

Methods:

1. **Grid Search:** Test all combinations (p,d,q)
2. **Stepwise Algorithm:** Sequential optimization
3. **Information Criteria:** AIC or BIC-based selection

### ### Information Criteria

#### #### AIC (Akaike Information Criterion)

$$AIC = 2k - 2\ln(L)$$

- k: number of parameters
- L: maximum likelihood
- Lower AIC = Better fit

#### #### BIC (Bayesian Information Criterion)

$$BIC = k \cdot \ln(n) - 2\ln(L)$$

- n: number of observations
- Penalizes complexity more than AIC
- More conservative model selection

**Recommendation:** Use AIC for forecast accuracy, BIC for parsimony

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## ## 5. Parameter Estimation

### ### Method of Estimation

## **\*\*Maximum Likelihood Estimation (MLE)\*\***

Finds parameters that maximize the probability of observed data.

Process:

1. Specify ARIMA(p,d,q) model
2. Calculate likelihood for all parameter values
3. Select parameters with highest likelihood

### **### Convergence Issues**

When model fails to converge:

- Try simpler model (lower p, q)
- Check data quality
- Increase differencing (d)
- Use different optimization algorithm

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## **## 6. Diagnostics and Validation**

### **### Residual Analysis**

ARIMA assumes residuals are:

- Independently distributed
- Normally distributed
- Have zero mean
- Have constant variance

### **### Tests**

#### **#### 1. Ljung-Box Test (Autocorrelation)**

Tests if residuals are uncorrelated.

**\*\*Null Hypothesis:\*\*** Residuals are independently distributed

**\*\*Result:\*\*** p-value > 0.05 → Good ✓

#### **#### 2. Shapiro-Wilk Test (Normality)**

Tests if residuals follow normal distribution.

**\*\*Null Hypothesis:\*\*** Residuals are normally distributed

**\*\*Result:\*\*** p-value > 0.05 → Good ✓

#### **#### 3. Augmented Dickey-Fuller Test (Stationarity)**

Tests if residuals are stationary.

**\*\*Result:\*\*** Stationary residuals → Good ✓

### ### Performance Metrics

#### #### RMSE (Root Mean Square Error)

$$\text{RMSE} = \sqrt{(\sum (y_t - \hat{y}_t)^2 / n)}$$

- Penalizes large errors
- In original units
- Lower is better

#### #### MAE (Mean Absolute Error)

$$\text{MAE} = \sum |y_t - \hat{y}_t| / n$$

- More robust to outliers than RMSE
- In original units
- Lower is better

#### #### MAPE (Mean Absolute Percentage Error)

$$\text{MAPE} = (\sum |y_t - \hat{y}_t| / |y_t|) / n \times 100\%$$

- Scale-independent
- Percentage error
- Better for comparing across series

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### ## 7. Forecasting with ARIMA

#### ### Point Forecasts

Forecasts at specific future time points.

**Notation:**  $\hat{y}_{t+h|t}$

Meaning: Forecast for time  $t+h$  based on data up to time  $t$

#### ### Confidence Intervals

**95% Confidence Interval:**

$$\hat{y}_{t+h} \pm 1.96 \times \text{SE}(\hat{y}_{t+h})$$

Where SE = standard error of forecast

**Properties:**

- Intervals widen as forecast horizon increases
- Reflects increasing uncertainty
- Smaller for stationary series

### Forecast Horizon

- **Short-term (1-10 periods):** High accuracy
- **Medium-term (10-30 periods):** Moderate accuracy
- **Long-term (30+ periods):** Low accuracy, wide intervals

### Forecast Evaluation

#### **In-Sample Accuracy:**

- Fit quality on training data
- May overestimate actual performance

#### **Out-of-Sample Accuracy:**

- Performance on test data (holdout)
- More reliable
- Recommended approach

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## 8. Practical Examples

### Example 1: NIFTY 50 Forecasting

**Data:** Daily closing prices, 250 observations

#### **Step 1: Stationarity Testing**

...

ADF Test: Statistic = -1.52, p-value = 0.52

KPSS Test: Statistic = 2.34, p-value < 0.05

Conclusion: Non-stationary, apply differencing

...

#### **Step 2: ACF/PACF Analysis**

...

After d=1:

ACF: Decays to zero gradually

PACF: Significant spike at lag 1

Suggestion: AR(1) or ARMA(1,1)

...

#### **Step 3: Model Fitting**

...

ARIMA(1,1,1):

AIC = 2456.23

BIC = 2468.45

RMSE = 45.32

...

#### **Step 4: Diagnostics**



...

Ljung-Box: p-value = 0.42 ✓

Shapiro-Wilk: p-value = 0.08 ✓

Conclusion: Model passes diagnostic tests

...

**\*\*Step 5: Forecast\*\***

...

20-day forecast:

Day 1: 21450 ± 120

Day 10: 21480 ± 420

Day 20: 21520 ± 650

...

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## ## Summary

**\*\*ARIMA Models:\*\***

- Combine autoregressive and moving average components
- Require stationary data
- Use (p,d,q) parameters
- Selected via information criteria or ACF/PACF analysis
- Generate point forecasts with confidence intervals

**\*\*Best Practices:\*\***

1. Test for stationarity
2. Apply minimum necessary differencing
3. Use information criteria for selection
4. Validate residual diagnostics
5. Evaluate on holdout test data
6. Report with confidence intervals

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**\*\*End of Document\*\***

For implementation examples, see USAGE.md and API\_REFERENCE.md

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